

2. Which planet is known as the 'Evening Star'?

- (a) Mars
- (b) Jupiter
- (c) Venus
- (d) Saturn

44th B.P.S.C. (Pre) 2000

Ans. (c)

Venus is known as the Evening Star and also as the Morning Star while Mars is called the Red Planet, and Saturn has the lowest density among all planets (lower than water also).

3. Which one of the following is known as 'Morning Star'?

- (a) Mercury
- (b) Venus
- (c) Mars
- (d) Saturn

U.P. Lower Sub. (Pre) 2015

Ans. (b)

Venus is called the 'Morning Star' and 'Evening Star' as it is the brightest planet of all. Venus is also called the 'twin sister' of Earth as Venus and Earth are almost the same size and mass and have a very similar composition.

4. Assertion (A) : Existence of human life on Venus is highly improbable.

Reason (R) : Venus has an extremely high level of carbon dioxide in its atmosphere.

Code :

- (a) Both A and R are true, and R is the correct explanation of A
- (b) Both A and R are true, and R is not the correct explanation of A
- (c) A is true, but R is false
- (d) A is false, but R is true

I.A.S. (Pre) 2005

Ans. (a)

There is evidence of the atmosphere on Venus. The atmosphere of Venus mainly contains carbon dioxide (CO₂) 96.5%. In addition to this, the surface temperature of Venus is more than 400° C. Hence due to the above reasons (CO₂) and high temperature, the possibility of life on Venus is highly unlikely. Hence, both Assertion (A) and Reason (R) are correct and (R) is the correct explanation of Assertion (A).

5. The planet which is called twin sister of the earth is –

- (a) Mercury
- (b) Venus
- (c) Mars
- (d) Pluto

U.P. Lower Sub. (Pre) 2008

U.P.P.C.S. (Pre) 2007

Ans. (b)

Venus is called the sister planet or twin planet of Earth. Venus is also known as the brightest planet, the Morning Star and the Evening Star etc.

6. The Spacecraft “Magellan” was sent to –

- (a) Pluto
- (b) Saturn
- (c) Mars
- (d) Venus

U.P.P.C.S. (Pre) 1991

Ans. (d)

The Magellan spacecraft was sent to Venus on a surface mapping mission. The spacecraft was launched on 04 May 1989, which arrived at Venus orbit on 10 August, 1990. It was named after Portuguese explorer of the sixteenth century, Ferdinand Magellan.

vi. The Earth

*The Earth is the third planet from the sun behind Mercury and Venus respectively. It is located between Venus and Mars. Survival of life is possible on our planet because of its ecology. Hence Earth has been given the name Green Planet. *Greek Philosopher Aristotle (384-322 B.C.) has clearly mentioned in second volume of the 14th chapter of his book "On The Heavens". But now it is universally accepted that the shape of our planet is much closer to being an **oblate spheroid**. Hence, the shape of our planet is called Geoid (Geoid meaning earth like). *Our planet Earth is tilted on its axis at an angle of **23°30'**. *The Earth takes 24 hours to rotate once on its axis. This leads to the phenomena of Day and Night. *Basically that part of the planet which is facing the sun experiences day and the opposite side experiences night. *The axial tilt of 23°30' (now 23°44') as it moves around the Sun causes seasonal changes. *The **moon** is an astronomical body that orbits planet Earth and is Earth's only permanent natural satellite. *On 21st March and 23rd September the Sun shines directly above the equator twice. From 21st March to 23rd September (6 months) the Sun shines between the equator and tropic of cancer. Thus, the Northern Hemisphere experiences summers and the Southern Hemisphere experiences winters. *On the other hand, the next 6 months i.e. from 23rd September to 21st of March the sun shines between the equator and the tropic of Capricorn. Hence, the southern hemisphere experiences summers and the northern hemisphere experiences winters. *Because of the slanted rays falling on the hemisphere, opposite to the one receiving direct sunlight, at any point north of the Arctic Circle

or south of the Antarctic circle there will be one period in the summer when the sun does not set, and one period in winter when the sun does not rise. For example in Northern Hemisphere the sun does not set at the North Pole and vice-versa.

*On June 21st, the sun is directly overhead the Tropic of Cancer. This phenomenon is known as the **Summer Solstice**. On 22nd December, the sun is directly overhead the Tropic of Capricorn. This phenomenon is called the **Winter Solstice**. *The 45-degree latitude is often called the halfway point between the equator and the North Pole. The cosine 45 latitude equals to 0.707, hence the length of this latitude is $0.707 \times 40090.2857 = 28343.832$ km. Therefore, the length of the 45° latitude is more than half of the length of the equator. The length of 60-degree latitude is half the length of the equator (cosine 60° = 0.5). The circumference of the equator is appx. 40090 km. Whereas the equatorial diameter of the Earth is appx. 12800 (12756 km). *The rotation of the Earth creates a **centrifugal force** which is **maximum at the equator**. As a result of this force, the weight of any object is minimum at the equator and increases with an increase in latitudes and is maximum at the poles. *The movement of the Earth in its orbit around the sun is called a revolution. It takes Earth **365.25 Days** or **365 Days, 6 hours 9 minutes and 9.51 seconds** to complete one revolution. *A **tropical year** (also known as a solar year) is the time that the Sun takes to return to the same position in the cycles of the seasons, as seen from Earth; for example, the time from the Vernal equinox to Vernal equinox or from summer solstice to summer solstice. *The one tropical year is approx. 365.24 days. Normally a calendar is based on a tropical year, which is 365 days, 5 hours, 48 min. and 45.2 seconds long. *The year 2000 AD witnessed an increase of 5 minutes in the length of the year. The length of the year was 365 days, 6 hours, 13 min. and 53.26 seconds. These types of years are called **Anomalistic years**. *The Earth revolves around the Sun at an average speed of **29.78 km/sec** (appx. 30km/sec). The revolution speed of the other planets in the solar system is as follows – (1) Mercury 47.4 km/s, (2) Venus – 35 km/s, (3) Earth – 29.78 km/s (4) Mars – 24.1 km/s (5) Jupiter – 13.1 km/s, (6) Saturn – 9.7 km/s (7) Uranus – 6.8 km/s (8) Neptune – 5.4 km/s. *The average distance between the sun and the Earth is 149597870.7 km. It takes appx. **8 min. and 19 seconds** for the Sun's rays to reach our planet. These electromagnetic energy waves travel at a speed of 186282 miles/sec. The closest star

to Earth is the Sun. The distance of other stars from our planet Earth is as follows– (1) Alpha Centauri - 4.35 light years, (2) Sun – 149.6×10^6 km, (3) Sirius A (Dog Star) – 8.6 light years.

*The Earth's magnetic field varies from place to place. Thumba is an ideal location for satellites as a result, The **Thumba Equatorial Rocket Launching Station** was established here. *Thumba is located in Thiruvananthapuram, the district of Kerala. *The projection of maximum artificial satellites is towards the east of the planet Earth. This is because a satellite launched towards the east direction will get an initial boost equal to the velocity of the earth surface. Thus, the initial boost helps in cutting down the cost of rockets used to launch the satellite. However, a satellite is not always projected towards the east. Such satellites that are to be placed in polar orbits are launched in the south or north directions.

*The inner core of our planet is solid. The outer core is in liquid form. The core has a depth of 2900 km. *The crust is composed of different elements in varied proportions. They are as follows (according to percentage by weight) - **Oxygen (O)** – 46.60%, **Silicon (Si)** – 27.72%, **Aluminium (Al)** – 8.13%, **Iron (Fe)** – 5.00%, **Calcium (Ca)** – 3.63%, **Sodium (Na)** - 2.83 %, **Potassium (K)** – 2.59%, **Magnesium (Mg)** – 2.09%. ***Earth Day** is an annual event celebrated on **April 22** worldwide. It was first observed in the year 1970.

1. Who amongst the following was the first to state that the Earth was spherical?

- | | |
|---------------|----------------|
| (a) Aristotle | (b) Copernicus |
| (c) Ptolemy | (d) Strabo |

I.A.S. (Pre) 2001

Ans. (a)

Aristotle (384-322 BC) was a Greek philosopher and polymath. In chapter -14, Volume-2 of his book “On the Heavens” he described Earth as a spherical body. This book was translated into English by J. L. Stocks. The geographer and philosopher Ptolemy (second century A.D.) Strabo (first century BC - First century AD.) and Copernicus (15th-16th century A.D.) were the geographer/philosopher after Aristotle. Thus Aristotle was the first philosopher who argued that the Earth is spherical in shape.

2. In the form of the organism, the Scientific name of Earth is –

- | | |
|------------------|------------|
| (a) Cow | (b) Sita |
| (c) Green Planet | (d) Harmij |

U.P.P.C.S. (Pre) 1999

Ans. (c)

The life on the Earth is possible due to the ecology present on it. Earth is known as the Green Planet for the presence of ecology.

3. The diameter of the Earth is –

- (a) 8,000 km
- (b) 1,00,000 km
- (c) 12,800 km
- (d) None of the above

Uttarakhand P.C.S. (Mains) 2006

Ans. (c)

The equatorial diameter of the Earth is 12,756 km (Near about 12800 km.) and the polar diameter is 12,714 km.

4. Statement (A) : Moving from one place on earth to another place of high latitude, the weight of an object decreases.

Reason (R) : Earth is not a perfect sphere.

Choose the correct answer by using code given below –

- (a) (A) and (R) both are true, and (R) explains (A) properly.
- (b) (A) and (R) both are true, but (R) does not explain (A) properly.
- (c) (A) is true, but (R) is wrong.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2003

Ans. (d)

The centrifugal force generated by the Earth's axis rotation is maximum at the equator. As a result of this force, the weight of any object is minimum at the Equator and maximum at poles and also increases towards high latitudes. Thus, statement (A) is false.

The shape of Earth is nearest to an oblate spheroid. The Earth is pudgier at the equator than at the poles due to Earth rotation. Earth is not perfectly spherical in shape and this shape is known as the geoid. Thus, Reason (R) is true.

5. Unlike the fluid core of the Earth, the core of the moon is–

- (a) Plasma
- (b) Volatile gas
- (c) Viscous liquid
- (d) Solid

U.P.P.C.S. (Mains) 2004

Ans. (d)

The fluid core of the Earth while the core of the moon is solid.

6. A chemical element found in abundance in the Earth's crust is–

- (a) Oxygen
- (b) Aluminium
- (c) Iron
- (d) Silicon

U. P.U.D.A./L.D.A. (Pre) 2013

Ans. (a)

Oxygen is the chemical element found in abundance in the Earth's crust. The percentage of various chemical elements in Earth's crust is as follows –

(Element)	(Percentage by weight)
Oxygen (O)	- 46.60
Silicon (Si)	- 27.72
Aluminium (Al)	- 8.13
Iron (Fe)	- 5.00
Calcium (Ca)	- 3.63
Sodium (Na)	- 2.83
Potassium (K)	- 2.59
Magnesium (Mg)	- 2.09

7. Time taken by light emitted from sun to reach earth is –

- (a) 2 minutes
- (b) 1 minutes
- (c) 8 minutes
- (d) 16 minutes

U.P.P.C.S. (Mains) 2005

M.P.P.C.S. (Pre) 1996

38th B.P.S.C. (Pre) 1992

Ans. (c)

The radiation from the sun which is popularly known as sunlight is a mixture of electromagnetic waves. Sunlight takes an average of 8 minutes and 19 seconds to travel from the Sun to the Earth. These electromagnetic energy waves travel with a speed of 3.0×10^{18} m./sec. Thus, the closest answer given in option (c) is true.

8. The sunlight from the sun to the earth reaches in :

- (a) 5 minutes
- (b) 6 minutes
- (c) 8 minutes
- (d) 10 minutes
- (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Light travels at a speed of 3.0×10^{18} m./sec. It takes 499.0 sec. (approx. 8 min 19 sec.) for light to travel from the sun to the Earth.

9. Which one of the following Stars is nearest to the Earth?

- (a) Polaris
- (b) Alpha Centauri
- (c) Sun
- (d) Sirius

I.A.S. (Pre) 1997

Ans. (c)

The Sun is the closest star to the Earth, about 149.6×10^6 kilometers away. Polaris is at a distance of 434 light years (new estimate 323 light years) from the Earth while Sirius A is at a distance of 8.6 light years from Earth. Alpha Centauri is the nearest neighbor after the sun, with a distance of 4.35 light years from the Earth.

10. Which of the following planet is nearest to earth ?

- (a) Pluto (b) Mars
(c) Saturn (d) Jupiter

M.P.P.C.S. (Pre) 1993

Ans. (b)

The list of planets in increasing order of distance from Earth is—

Venus	- 41.4 Million km
Mars	- 78.4 Million km
Mercury	- 91.7 Million km
Jupiter	- 628.9 Million km
Saturn	- 1282.4 Million km
Uranus	- 2717.4 Million km
Neptune	- 4365.4 Million km

As the planet Venus is not given in the options, so the correct answer in the given option is (b) i.e. Mars.

11. The planet nearest to the Earth is –

- (a) Venus (b) Mercury
(c) Mars (d) Jupiter

U.P.P.C.S. (Pre) 1991

U.P.P.C.S. (Pre) 1993

U.P.P.C.S. (Pre) 1997

Uttarakhand P.C.S. (Pre) 2002

U.P.P.C.S. (Pre) 2012

Ans. (a)

See the explanation of the above question.

12. In completing one revolution of the sun, Earth takes approximately –

- (a) 365 days (b) 365.25 days
(c) 365.5 days (d) 365.75 days

41st B.P.S.C. (Pre) 1996

Ans. (b)

The Earth takes 365.25636 days or 365 days, 6 hours, 9 minutes and 9.51 second to complete one orbit around the Sun. This period is known as 'Solar year'. The solar year is also known as The tropical year. The tropical year is equal to 365.24 days. In the year 2000, an increase of 5 minute had been noted in comparison to other years. In this period the time was calculated to 365 days, 6 hours, 13 minutes (in place of 9 minutes), 53.26 seconds (in place of 9.51 seconds). Such a year is known as the 'Anomalistic Year'.

13. At which average speed the Earth revolves around the Sun?

- (a) 20 km/ sec (b) 30 km/ sec
(c) 40 km/ sec (d) 50 km/ sec

I.A.S. (Pre) 2006

Ans. (b)

Earth revolves around the Sun with an average speed (orbital velocity) of 29.8 km/second (30 km/sec). The orbital velocity is 29.8 km/ sec. This speed is comparative to the speed of light. The orbital velocity of other planets is as follows –

- (1) Mercury - 47.4 km/ sec
(2) Venus - 35 km/ sec
(3) Earth - 29.8 km/ sec
(4) Mars - 24.1 km/ sec
(5) Jupiter - 13.1 km/ sec
(6) Saturn - 9.7 km/ sec
(7) Uranus - 6.8 km/ sec
(8) Neptune - 5.4 km/ sec

14. Approximately how much distance per minute does earth cover while revolving?

- (a) 49 km (b) 59 km
(c) 69 km (d) 79 km

44th B.P.S.C. (Pre) 2001

Ans. (*)

Earth's equatorial perimeter is about 40,090.28 km. This distance is covered by Earth in 24 hours and this indicates that the cruising speed of Earth on its axis per hour is 40,090.28 km/ 24 hours = 1670 km/ hour (approx). According to this calculation the cruising speed per minute is = 1670 km/ 60 min = 27.8 km/ min. Thus, none of the given answers is correct. Note - In this question rotating should be in place of revolving.

15. The equatorial circumference of the Earth is nearly

- (a) 31,000 km (b) 40,000 km
(c) 50,000 km (d) 64,000 km

Uttarakhand P.C.S. (Pre) 2016

Ans. (b)

See the explanation of the above question.

16. The Earth's axis of rotation (polar axis) is always inclined at an angle of

- (a) 23.0° from the elliptic axis
(b) 23.1° from the elliptic axis
(c) 24.5° from the elliptic axis
(d) 23.5° from the elliptic axis

U.P.P.C.S. (Pre) 2016

Ans. (d)

The Earth rotates about an imaginary line that passes through the North and South poles of the planet. This line is called the axis of rotation. The Earth axis of rotation is always inclined at an angle of 23°30' (23.5°) from the elliptic axis.

17. Consider the following statements :

1. The axis of the earth's magnetic field is inclined at 23 and a half degrees to the geographic axis of the earth.
2. The earth's magnetic pole in the northern

hemisphere is located on a peninsula in northern Canada.

3. Earth's magnetic equator passes through Thumba in South India.

Which of the Statements given above is/are correct?

- (a) 1, 2 and 3 (b) 2 and 3
(c) 2 only (d) 3 only

I.A.S. (Pre) 2005

Ans. (d)

The axis of the magnetic field of Earth is different at different places, not equally inclined at $23\frac{1}{2}^\circ$. Thus, statement 1 is wrong. Earlier the magnetic pole of the Earth in the northern hemisphere was situated at Queen Elizabeth Island of Northern Canada not on a peninsula, thus, statement (2) is also incorrect. Thumba is an ideal location for satellite launches because of its proximity to the geomagnetic equator. This is the reason for establishing Thumba Equatorial Orbit Launching Station (TELOS) here as it proves to be a suitable place for satellite launch. Thus statement 3 is true, and option (d) is correct.

18. Consider the following statements and select the correct answer from the codes given below:

- The northern end of the earth's axis is called North Pole.
- 45° latitude is half of the length of the equator.
- There is parallelism of the earth's axis.
- The earth's speed of revolution is faster in aphelion position.

Code :

- (a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 3

U.P.P.C.S. (Pre) 2000

Ans. (d)

The North Pole is the northernmost point of Earth i.e. northern end of the Earth's axis is called North Pole. So, statement (1) is true. 45° latitude passes from the middle of the equator and poles. The length of the 45° parallel line is 2×3333 km, while the length of the equator is 40000 km and so the 45° parallel is not half of the length of the equator and statement (2) is wrong. Parallelism of the Earth's axis means that as Earth revolves around the sun, the Earth's axis remains parallel to its previous positions. The situation of aphelion occurs on July, when the distance between the sun and Earth is maximum and in the situation of aphelion the cruising speed of Earth goes slow. Thus, statement (4) is wrong.

19. The causes of Day and Night is –

- (a) Earth's rotation
(b) Rotation of Earth on its axis
(c) Tilt of Earth's axis
(d) Moon's rotation

R.A.S./R.T.S. (Pre) 2000

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

The Earth experiences day and night as a result of Earth's rotation on its axis. Earth takes approximately 24 hours to complete one rotation. When a place on Earth faces the sun it has daytime. When it faces away from the sun it has night.

20. When the length of day and night is equal the Sunrays falls perpendicular at –

- (a) North pole (b) Equator
(c) South pole (d) Tropic of cancer

Uttarakhand P.C.S. (Pre) 2002

Ans. (b)

The sunrays fall perpendicular at the equator twice in a year (on 21 March and on 23 September). On these dates, the duration of day and night becomes an equal 12 hours, in both hemispheres.

21. Seasonal changes are caused by which one of the special characteristic of Earth's motion ?

- (a) Revolution around sun
(b) Axial tilt by $23\frac{1}{2}^\circ$ degree
(c) (a) and (b) both
(d) Rotation on its axis

R.A.S./R.T.S. (Pre) 1992

Ans. (c)

The biggest reason for seasonal changes is Earth's axial tilt by $23\frac{1}{2}^\circ$ degrees and Earth's revolution around the sun. The variability of the distance of Earth from sun has also slight importance in regard to seasonal variability.

22. What causes the change of Seasons?

- (a) Earth's rotation and revolution
(b) Earth's revolution
(c) Earth's revolution and inclination of its axis
(d) Earth's rotation and inclination of its axis

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

See the explanation of the above question.

23. Consider one special point at the Earth's surface (eg. City Delhi) its temperature at noon (eg. 12 O'clock) will be higher in Summer in comparison to winter because-

- (a) Earth is much closer to Sun in Summer in comparison to winter.
- (b) In winter the Sun rays falling on the Earth bends more towards the Earth's surface in Comparison to the summer.
- (c) The vaporization of water happens only in winter due to environmental precipitation
- (d) In winters the Earth's axis becomes more inclined towards Sun.

I.A.S. (Pre) 1993

Ans. (b)

Twice in a year on 21st March and 23rd September, Sun rays fall perpendicular to the Earth at the Equator. From 21st March to 23rd September, the Sun stands between the Equator and Tropic of Cancer. In the same period, the summer season occurs in Northern Hemisphere and winter occurs in the Southern Hemisphere. The sunrays fall in askew position in Southern Hemisphere than in Northern Hemisphere, while opposite to it from 23rd September to 21st March the Sun stands between the Equator and Tropic of Capricorn. In this period, the Sun's rays fall in a more askew position in Northern Hemisphere than in Southern Hemisphere and summer occurs in Southern Hemisphere and winter occurs in Northern Hemisphere. Due to this process, the temperature of Delhi, situated in North the Hemisphere; remains higher in summer season as compared to the winter season.

24. If the distance between the sun and the earth is reduced by quarter to present distance, then probability of which one will increase –

- (a) Earth would fall into Sun.
- (b) Earth shall be burned.
- (c) The duration of our year will be decreased.
- (d) Earth will fly into space.

R.A.S./R.T.S. (Pre) 1993

Ans. (c)

If the distance between the Sun and Earth is reduced by a quarter of the present distance, then the size of Earth's orbit in which it revolves around the Sun will be smaller and consequently the duration of the year would be less than 365.25 days.
Due to the reduction in the present distance, the speed with which Earth is revolving will increase.

25. Assertion (A) : Artificial satellites are always launched from the earth in the eastward direction.

Reason (R) : The earth rotates from west to east and so the Satellite attains the escape velocity.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, (R) is not a correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2002

Ans. (d)

It is true that most satellites are launched in the east direction from Earth because satellites gain momentum but it is not true for every satellites launch. Satellite to orbit poles are launched in the direction of north or south.

26. A Ball thrown outside from an artificial satellite revolving around the Earth will –

- (a) Go to the Sun
- (b) Go to the Moon
- (c) Fall on the Earth
- (d) Revolve around the Earth in the same orbit with the same time period as satellite

U.P.P.C.S. (Pre) 1995

Ans. (d)

Any object thrown out of artificial satellite will revolve around like the satellite.

27. Which elements are abundant in the formation of interior layer of the earth?

- (a) Silica and aluminium
- (b) Silica and magnesium
- (c) Basalt and silica
- (d) Nickel and ferrum
- (e) None of the above / More than one of the above.

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

The elements nickel and ferrum are abundant in the formation of Nife, the inner layer of the Earth. Nife is an abbreviation of Nickel and Ferrum (Iron).

28. In the structure of planet Earth, below the mantle, the core is mainly made up of which one of the following ?

- (a) Aluminium
- (b) Chromium

(c) Iron

(d) Silicon

I.A.S. (Pre) 2009

Ans. (c)

There is an outer core and inner core below the mantle in the structure of the Earth. The outer core is about 2260 km thick (radius) and mostly composed of liquid iron and nickel. The inner core is a hot dense ball of (mostly) iron. It has a radius of about 1220 km.

29. The World Earth Day is celebrated on –

(a) 22 April

(b) 22 March

(c) 07 April

(d) 07 March

M.P.P.C.S. (Pre) 2015

Ans. (a)

Earth Day is celebrated on 22nd April every year to mark the birth anniversary of the modern environmental movement in 1970. It was the first time celebrated in the year 1970.

30. When is the 'International Earth Day' celebrated?

(a) 20th April

(b) 5th June

(c) 22nd April

(d) 3rd March

M.P. P.C.S. (Pre) 2018

Ans. (c)

See the explanation of the above question.

31. Earth's axis is tilted at an angle of –

(a) 23 1/2°

(b) 66 1/2°

(c) 33 1/2°

(d) 42 1/2°

M.P.P.C.S. (Pre) 1990

Ans. (a)

Earth is tilted at an angle of $23\frac{1}{2}^{\circ}$ (now 23.44°) on its axis. It rotates around on its axis with approximately a speed of 1037 miles per hour. Earth takes approximately 24 hours to complete one rotation on its axis.

32. Consider the following statements :

1. The Earth's magnetic field has reversed every few hundred thousand years.
2. When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide
3. When living organisms originated, they modified the early atmosphere of the Earth.

Which of the statements given above is/are correct?

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (c)

According to a data published by the European Space Agency, the Earth's magnetic field could flip. Something causes the movement of the molten core to change resulting in the reversal of the North and South magnetic poles.

Earth is approximately 4.6 billion years old. The earlier environment had an absence of free oxygen. Around 4 billion years ago water vapor present in the atmosphere condensed to form the hydrosphere. Around 1 billion years ago early aquatic animals like algae came into existence. They Being autotrophs, they started the process of photosynthesis which led to the release of free oxygen in the atmosphere. In the upper layers of the atmosphere this free oxygen led to the formation of ozone.

33. Who among the following was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the Sun?

(a) Aryabhatta

(b) Bhaskara

(c) Brahmagupta

(d) Varahamihira

(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Aryabhatta was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the sun. His works are compiled in Aryasiddhanta. Bhaskara-I was a proponent of the Bhedabheda School of Vedanta Philosophy. Varahamihira's notable works include Brihat Samhita and Pancha Siddhantika.

34. The possibility of a desert on earth is more

(a) Nearby 0° latitude

(b) Nearby 23° latitude

(c) Nearby 50° latitude

(d) Nearby 70° latitude

41st B.P.S.C. (Pre) 1996

Ans. (b)

Desert climate is also known as arid climate. The perfect climate for the desert is tropical climate which is spread mostly between 15-30° latitude. There are four main geographical situations for the formation of deserts –

- (1) The presence of high air pressure near 30° latitude. Sahara and Australian deserts fall under this climate.
- (2) The western coastal side of continents found between 20°-30° latitude. Arizona desert, Mexican desert, Atacama desert and Namibian desert come under this climate.
- (3) Death Valley (North America), Patagonian desert (Argentina) and Peruvian desert are the deserts found in mountains rainshadow region.
- (4) The deserts situated in the mid part of the continents, far from the humid air. Gobi desert (China and Mongolia), Australian desert and Great Basin desert (USA) come under this group.